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**Kelas : IF-1**

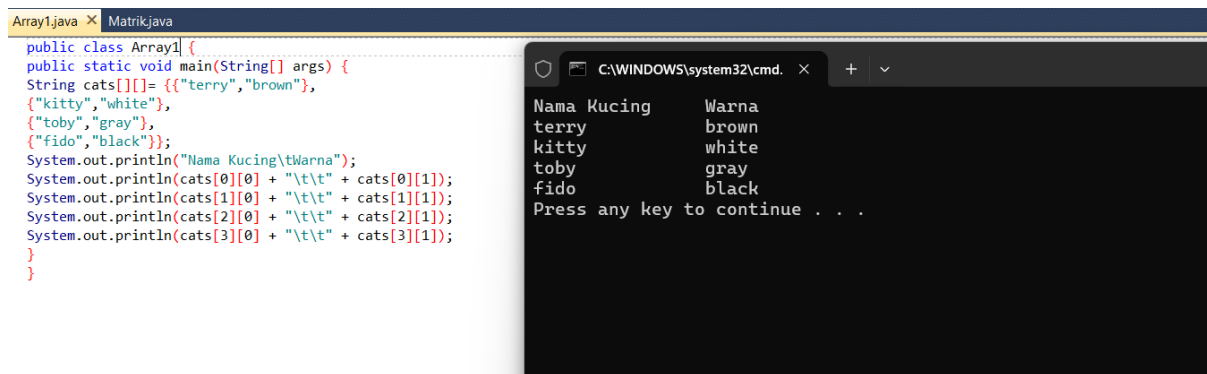
## **LISTING PRAKTIKUM ALGORITMA DAN PEMROGRAMAN LANJUT**

### **MODUL 4**

### **ARRAY 2 DIMENSI**

#### **PRAKTIK**

##### **Praktik 1**

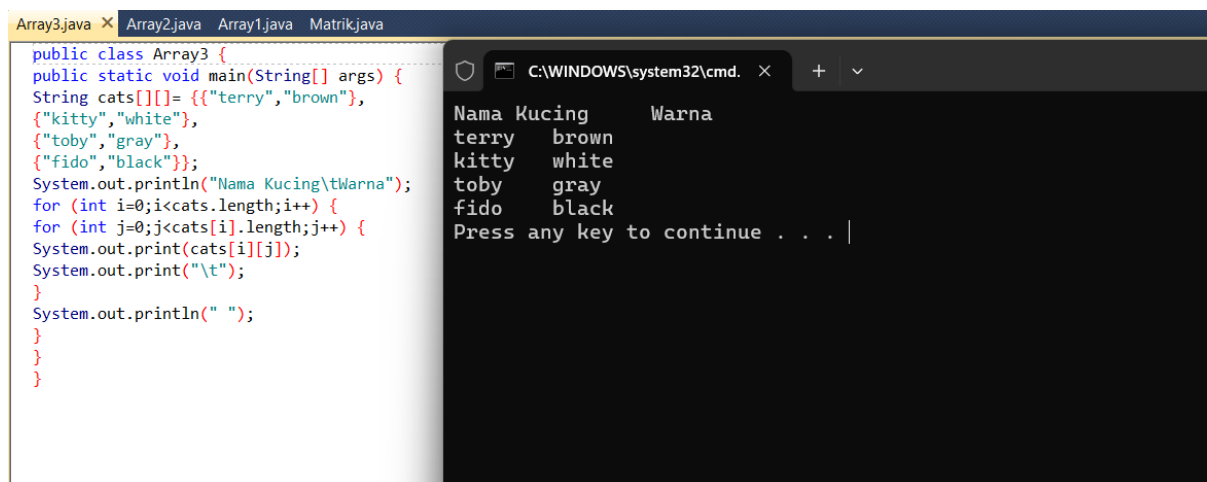


The screenshot shows a Java IDE with a file named `Array1.java`. The code defines a 2D array `cats` with 4 rows and 2 columns, containing cat names and colors. It prints the array using `System.out.println` with formatted tabs. To the right, a command prompt window shows the output of the program.

```
public class Array1 {  
    public static void main(String[] args) {  
        String cats[][]= {{ "terry", "brown"},  
        {"kitty", "white"},  
        {"toby", "gray"},  
        {"fido", "black"}}};  
        System.out.println("Nama Kucing\tWarna");  
        System.out.println(cats[0][0] + "\t\t" + cats[0][1]);  
        System.out.println(cats[1][0] + "\t\t" + cats[1][1]);  
        System.out.println(cats[2][0] + "\t\t" + cats[2][1]);  
        System.out.println(cats[3][0] + "\t\t" + cats[3][1]);  
    }  
}
```

```
Nama Kucing      Warna  
terry           brown  
kitty           white  
toby            gray  
fido            black  
Press any key to continue . . .
```

##### **Praktik 2**

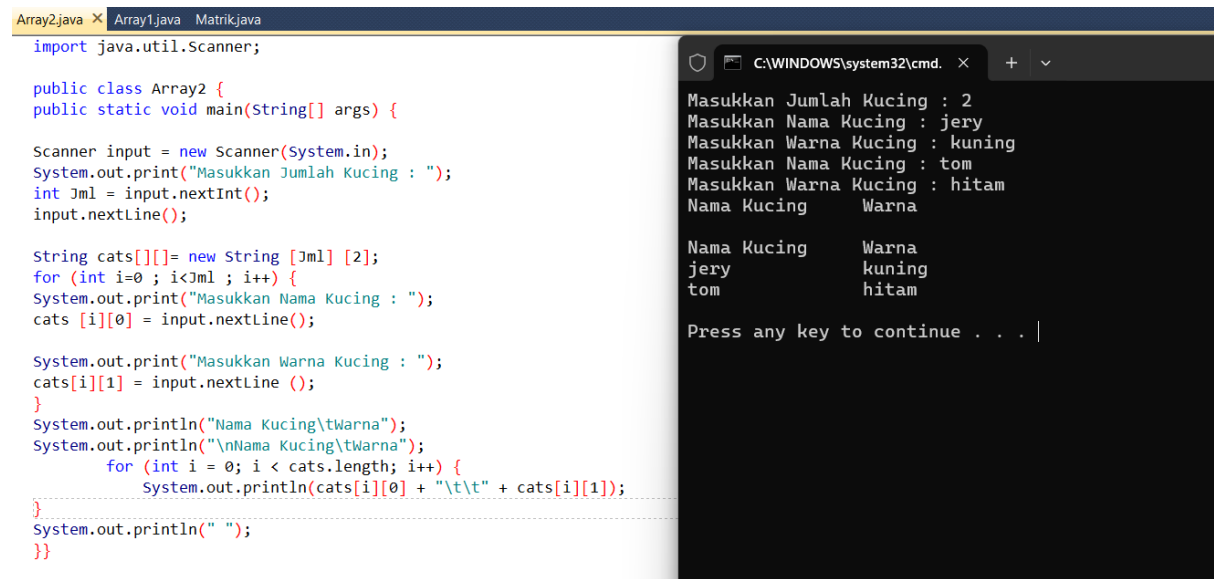


The screenshot shows a Java IDE with a file named `Array3.java`. The code defines a 2D array `cats` with 4 rows and 2 columns. It uses nested `for` loops to iterate through the array and print each element. To the right, a command prompt window shows the output of the program.

```
public class Array3 {  
    public static void main(String[] args) {  
        String cats[][]= {{ "terry", "brown"},  
        {"kitty", "white"},  
        {"toby", "gray"},  
        {"fido", "black"}}};  
        System.out.println("Nama Kucing\tWarna");  
        for (int i=0;i<cats.length;i++) {  
            for (int j=0;j<cats[i].length;j++) {  
                System.out.print(cats[i][j]);  
                System.out.print("\t");  
            }  
            System.out.println(" ");  
        }  
    }  
}
```

```
Nama Kucing      Warna  
terry    brown  
kitty    white  
toby     gray  
fido     black  
Press any key to continue . . . |
```

## Praktik 3



```
import java.util.Scanner;

public class Array2 {
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        System.out.print("Masukkan Jumlah Kucing : ");
        int Jml = input.nextInt();
        input.nextLine();

        String cats[][]= new String [Jml] [2];
        for (int i=0 ; i<Jml ; i++) {
            System.out.print("Masukkan Nama Kucing : ");
            cats [i][0] = input.nextLine();

            System.out.print("Masukkan Warna Kucing : ");
            cats[i][1] = input.nextLine ();
        }
        System.out.println("Nama Kucing\tWarna");
        System.out.println("\nNama Kucing\tWarna");
        for (int i = 0; i < cats.length; i++) {
            System.out.println(cats[i][0] + "\t\t" + cats[i][1]);
        }
        System.out.println(" ");
    }
}
```

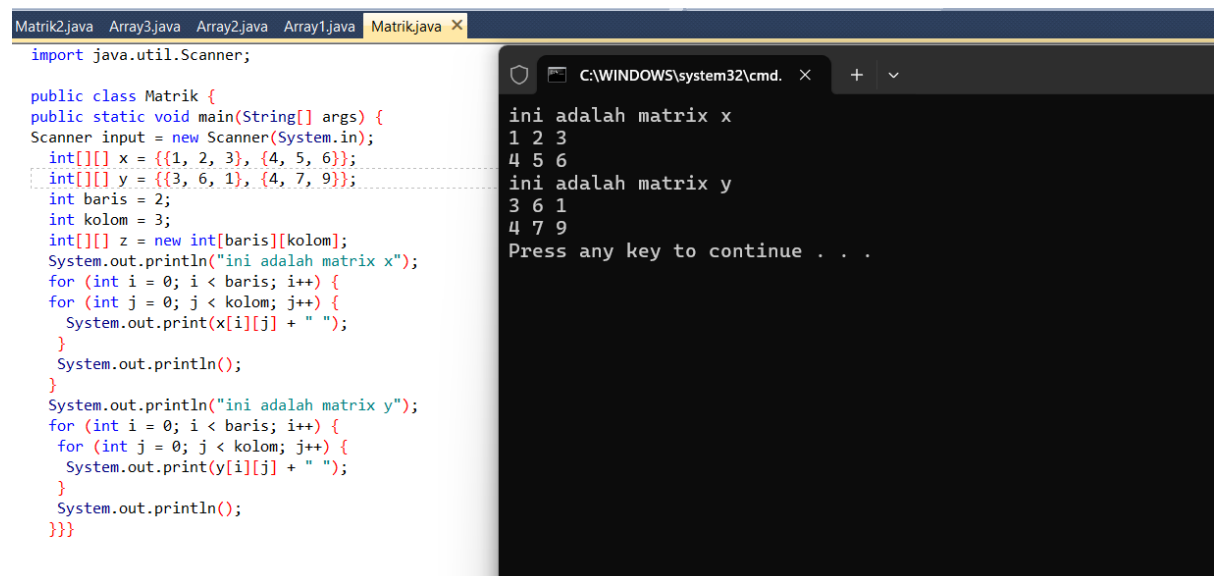
C:\WINDOWS\system32\cmd. x + v

Masukkan Jumlah Kucing : 2  
Masukkan Nama Kucing : jery  
Masukkan Warna Kucing : kuning  
Masukkan Nama Kucing : tom  
Masukkan Warna Kucing : hitam  
Nama Kucing      Warna

| Nama Kucing | Warna  |
|-------------|--------|
| jery        | kuning |
| tom         | hitam  |

Press any key to continue . . .

## Praktik 4



```
import java.util.Scanner;

public class Matrik {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[][] x = {{1, 2, 3}, {4, 5, 6}};
        int[][] y = {{3, 6, 1}, {4, 7, 9}};
        int baris = 2;
        int kolom = 3;
        int[][] z = new int[baris][kolom];
        System.out.println("ini adalah matrix x");
        for (int i = 0; i < baris; i++) {
            for (int j = 0; j < kolom; j++) {
                System.out.print(x[i][j] + " ");
            }
            System.out.println();
        }
        System.out.println("ini adalah matrix y");
        for (int i = 0; i < baris; i++) {
            for (int j = 0; j < kolom; j++) {
                System.out.print(y[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

C:\WINDOWS\system32\cmd. x + v

ini adalah matrix x  
1 2 3  
4 5 6  
ini adalah matrix y  
3 6 1  
4 7 9  
Press any key to continue . . .

## Praktik 5

```
MatrikTranspose.java Matrik2.java X Array3.java Array2.java Array1.java Matrik.java

import java.util.Scanner;
public class Matrik2 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[][] x = {{1, 2, 3}, {4, 5, 6}};
        int[][] y = {{3, 6, 1}, {4, 7, 9}};
        int baris = 2;
        int kolom = 3;
        int[][] hasilTambah = new int[baris][kolom];
        int[][] hasilKurang = new int[baris][kolom];
        System.out.println("Ini adalah matrix X:");
        for (int i = 0; i < baris; i++) {
            for (int j = 0; j < kolom; j++) {
                System.out.print(x[i][j] + " ");
            }
            System.out.println();
        }
        System.out.println("Ini adalah matrix Y:");
        for (int i = 0; i < baris; i++) {
            for (int j = 0; j < kolom; j++) {
                System.out.print(y[i][j] + " ");
            }
            System.out.println();
        }
        for (int i = 0; i < baris; i++) {
            for (int j = 0; j < kolom; j++) {
                hasilTambah[i][j] = x[i][j] + y[i][j];
                hasilKurang[i][j] = x[i][j] - y[i][j];
            }
        }
        System.out.println("Hasil Penjumlahan Matriks X + Y:");
        for (int i = 0; i < baris; i++) {
            for (int j = 0; j < kolom; j++) {
                System.out.print(hasilTambah[i][j] + " ");
            }
            System.out.println();
        }
        System.out.println("Hasil Pengurangan Matriks X - Y:");
        for (int i = 0; i < baris; i++) {
            for (int j = 0; j < kolom; j++) {
                System.out.print(hasilKurang[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

```
C:\WINDOWS\system32\cmd. X + v

Ini adalah matrix X:
1 2 3
4 5 6
Ini adalah matrix Y:
3 6 1
4 7 9
Hasil Penjumlahan Matriks X + Y:
4 8 4
8 12 15
Hasil Pengurangan Matriks X - Y:
-2 -4 2
0 -2 -3
Press any key to continue . . .
```

## Praktik 6

```
MatrikTranspose.java X Matrik2.java Array3.java Array2.java Array1.java Matrik.java

public class MatrikTranspose {
    public static void main(String[] args) {
        int[][] matriks = {{12,23,32},{34,56,63},{78,89,97}};
        int j,k;
        System.out.println("Matriks Sebelum Transpose ");
        for(j=0;j<3;j++){
            for(k=0;k<3;k++){
                System.out.print(matriks[j][k]+" ");
            }
            System.out.println();
        }
        System.out.println("\nMatriks Setelah Transpose");
        for(j=0;j<3;j++){
            for(k=0;k<3;k++){
                System.out.print(matriks[k][j]+" ");
            }
            System.out.println();
        }
    }
}
```

```
C:\WINDOWS\system32\cmd. X + v

Matriks Sebelum Transpose
12 23 32
34 56 63
78 89 97

Matriks Setelah Transpose
12 34 78
23 56 89
32 63 97
Press any key to continue . . .
```

## Praktik 7

```
Array2Dimensi1.java × Document3 Matrik.java JavaApplication3.java * JavaApplication2.java

import java.util.Scanner;
public class Array2Dimensi1 {
    public static void main(String[] args){
        int b=3;int d=3; //matrik 3 baris 3 kolom
        System.out.println("Masukan Nilai Matrix:");
        int[][] matrix1=new int[b][d];
        for(int i=0;i<b;i++){
            for(int j=0;j<d;j++){
                matrix1[i][j]=input();
            }
        }
        for(int i=0;i<b;i++){
            for(int j=0;j<d;j++){
                System.out.print(matrix1[i][j]+" ");
            }
            System.out.println();
        }
    }
    static int input(){
        Scanner a=new Scanner(System.in);
        int b=a.nextInt();
        return b;
    }
}
```

```
C:\WINDOWS\system32\cmd. × + v

Masukan Nilai Matrix:
1
2
3
4
5
6
7
8
9
1 2 3
4 5 6
7 8 9
Press any key to continue . . . |
```

## Praktik 8

```
Array2Dimensi3.java × Array2Dimensi2.java MatrikTranspose.java Matrik2.java Array3.java Array2.java Array1.java Matrik.java

import java.util.Scanner;
public class Array2Dimensi3 {
    public static void main(String[] args){
        int b=3; int d=3; // matriks 3 baris 3 kolom
        int[][] matrix1 = new int[b][d];
        int[][] matrix2 = new int[b][d];
        int[][] hasil = new int[b][d];

        System.out.println("Masukan Nilai Matrix 1:");
        for(int i=0;i<b;i++){
            for(int j=0;j<d;j++){
                matrix1[i][j]=input();
            }
        }
        System.out.println("Masukan Nilai Matrix 2:");
        for(int i=0;i<b;i++){
            for(int j=0;j<d;j++){
                matrix2[i][j]=input();
            }
        }
        for(int j=0;j<d;j++){
            hasil[i][j] = 0;
            for(int k=0;k<d;k++){
                hasil[i][j] += matrix1[i][k] * matrix2[k][j];
            }
        }
        for(int i=0;i<b;i++){
            for(int j=0;j<d;j++){
                System.out.print(hasil[i][j]+" ");
            }
            System.out.println();
        }
    }
    static int input(){
        Scanner a=new Scanner(System.in);
        int b=a.nextInt();
        return b;
    }
}
```

```
C:\WINDOWS\system32\cmd. × + v

Masukan Nilai Matrix 1:
1
2
3
4
5
6
7
8
9
Masukan Nilai Matrix 2:
1
2
3
4
5
6
7
8
9
30 36 42
66 81 96
102 126 150
Press any key to continue . . . |
```

# LATIHAN

## Latihan 1

```
Document2 JavaApplication2.java * X
import java.util.Scanner;
public class JavaApplication2 {
    public static void main(String[] args) throws Exception {
        Scanner input = new Scanner(System.in);
        int mhs,jml, banyakTes = 3, nilai[1][], ntt[], ntr[];
        float rata[], jumlah[], rtt, rtr;
        System.out.print("Masukkan Jumlah Mahasiswa : ");
        mhs = input.nextInt();
        nilai = new int[mhs][banyakTes];
        jumlah = new float[mhs];
        rata = new float[mhs];
        ntt = new int[banyakTes];
        ntr = new int[banyakTes];
        System.out.println();
        for(int h=0;h<mhs;h++){ //Mahasiswa
            System.out.println("Mahasiswa " + (h+1));
            for(int i=0;i<banyakTes;i++){ //Tes Keberapa
                System.out.print("Nilai Tes " + (i+1) + " : ");
                nilai[h][i] = input.nextInt();
                jumlah[h] = jumlah[h] + nilai[h][i];
            }
            rata[h] = jumlah[h]/banyakTes;
            System.out.println();
        }
        for(int i=0;i<banyakTes;i++){
            ntt[i] = nilai[0][i];
            ntr[i] = nilai[0][i];
        }
        rtt = rata[0];
        rtr = rata[0];
        for(int i=0;i<banyakTes;i++){
            for(int j=0;j<mhs;j++){
                if(ntt[i] < nilai[j][i]){
                    ntt[i] = nilai[j][i];
                }
                if(ntr[i] > nilai[j][i]){
                    ntr[i] = nilai[j][i];
                }
            }
        }
    }
}
```

```
C:\WINDOWS\system32\cmd. X + v
Masukkan Jumlah Mahasiswa : 2

Mahasiswa 1
Nilai Tes 1 : 90
Nilai Tes 2 : 50
Nilai Tes 3 : 90

Mahasiswa 2
Nilai Tes 1 : 90
Nilai Tes 2 : 78
Nilai Tes 3 : 95

-----
Daftar Nilai Mahasiswa :
-----

          Test 1  Test 2  Test 3  Rata-rata
Mahasiswa 1    90     50     90     76.666664
Mahasiswa 2    90     78     95     87.666664

Nilai Tertinggi 90     78     95     87.666664
Nilai Terendah  90     50     90     76.666664
Press any key to continue . . . |
```

```
Document2 JavaApplication2.java * X
Document2
for(int i=0;i<mhs;i++){
    if(rtt < rata[i]){
        rtt = rata[i];
    }
    if(rtr > rata[i]){
        rtr = rata[i];
    }
}
System.out.println("-----");
System.out.println("Daftar Nilai Mahasiswa : ");
System.out.println("-----");
System.out.println();
System.out.println("\t\tTest 1\tTest 2\tTest 3\tRata-rata");
for(int j=0;j<mhs;j++){
    System.out.print("Mahasiswa " + (j+1));
    for(int k=0;k<banyakTes;k++){
        System.out.print("\t" + nilai[j][k]);
    }
    System.out.print("\t" + rata[j]);
    System.out.println();
}
System.out.println();
System.out.print("Nilai Tertinggi\t");
for(int j=0;j<banyakTes;j++){//Nilai tertinggi
    System.out.print(ntt[j] + "\t");
}
System.out.print(rtt);//Rata-rata tertinggi
System.out.println();
System.out.print("Nilai Terendah\t");
for(int j=0;j<banyakTes;j++){//Nilai terendah
    System.out.print(ntr[j] + "\t");
}
System.out.print(rtr);//Rata-rata terendah
System.out.println();
}
```

## Latihan 2

```
JavaApplication3.java * JavaApplication2.java
import java.util.Scanner;

public class JavaApplication3 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int mhs, banyakTes = 3;
        int[][] nilai;
        float[] rata, jumlah;
        int[] ntt, ntr;
        float rtt, rtr;
        String[] nama, nim, jurusan;

        System.out.print("Masukkan Jumlah Mahasiswa : ");
        mhs = input.nextInt();
        input.nextLine();

        nilai = new int[mhs][banyakTes];
        jumlah = new float[mhs];
        rata = new float[mhs];
        ntt = new int[banyakTes];
        ntr = new int[banyakTes];
        nama = new String[mhs];
        nim = new String[mhs];
        jurusan = new String[mhs];
        System.out.println();

        for (int h = 0; h < mhs; h++) {
            System.out.println("Data Mahasiswa ke-" + (h + 1));
            System.out.print("Nama      : ");
            nama[h] = input.nextLine();
            System.out.print("NIM       : ");
            nim[h] = input.nextLine();
            System.out.print("Jurusan   : ");
            jurusan[h] = input.nextLine();

            System.out.println("Masukkan Nilai Mahasiswa : ");

            for (int i = 0; i < banyakTes; i++) {
                System.out.print("Nilai Tes " + (i + 1) + " : ");
                nilai[h][i] = input.nextInt();
                jumlah[h] += nilai[h][i];
            }

            rata[h] = jumlah[h] / banyakTes;
            input.nextLine();

            for (int i = 0; i < banyakTes; i++) {
                ntt[i] = nilai[h][i];
                ntr[i] = nilai[h][i];
            }

            rtt = rata[h];
            rtr = rata[h];

            for (int j = 0; j < mhs; j++) {
                if (ntt[i] < nilai[j][i]) {
                    ntt[i] = nilai[j][i];
                }
                if (ntr[i] > nilai[j][i]) {
                    ntr[i] = nilai[j][i];
                }
            }
        }

        System.out.println("Daftar Nilai Mahasiswa :");

        System.out.println();

        System.out.println("Mahasiswa 1");
        System.out.println("Nilai Tes 1 : 90");
        System.out.println("Nilai Tes 2 : 50");
        System.out.println("Nilai Tes 3 : 90");

        System.out.println("Mahasiswa 2");
        System.out.println("Nilai Tes 1 : 90");
        System.out.println("Nilai Tes 2 : 78");
        System.out.println("Nilai Tes 3 : 95");

        System.out.println();

        System.out.println("Daftar Nilai Mahasiswa :");

        System.out.println();

        System.out.println("Mahasiswa 1      Test 1 Test 2 Test 3 Rata-rata");
        System.out.println("Mahasiswa 1      90    50    90    76.666664");
        System.out.println("Mahasiswa 2      90    78    95    87.666664");

        System.out.println("Nilai Tertinggi 90    78    95    87.666664");
        System.out.println("Nilai Terendah  90    50    90    76.666664");
        System.out.println("Press any key to continue . . . |");
    }
}
```

```
System.out.println("Masukkan Nilai:");
for (int i = 0; i < banyakTes; i++) {
    System.out.print("Nilai Tes " + (i + 1) + " : ");
    nilai[h][i] = input.nextInt();
    jumlah[h] += nilai[h][i];
}
rata[h] = jumlah[h] / banyakTes;
input.nextLine();
System.out.println();

for (int i = 0; i < banyakTes; i++) {
    ntt[i] = nilai[0][i];
    ntr[i] = nilai[0][i];
}
rtt = rata[0];
rtr = rata[0];

for (int i = 0; i < banyakTes; i++) {
    for (int j = 0; j < mhs; j++) {
        if (ntt[i] < nilai[j][i]) {
            ntt[i] = nilai[j][i];
        }
        if (ntr[i] > nilai[j][i]) {
            ntr[i] = nilai[j][i];
        }
    }
}
```

```

for (int i = 0; i < mhs; i++) {
    if (rtt < rata[i]) rtt = rata[i];
    if (rtr > rata[i]) rtr = rata[i];
}

System.out.println("-----");
System.out.println("Daftar Nilai Mahasiswa : ");
System.out.println("-----\n");
System.out.println("Nama\t\tNIM\t\tJurusan\t\tTest1\tTest2\tTest3\tRata-rata");
for (int j = 0; j < mhs; j++) {
    System.out.print(nama[j] + "\t" + nim[j] + "\t" + jurusan[j] + "\t");
    for (int k = 0; k < banyakTes; k++) {
        System.out.print(nilai[j][k] + "\t");
    }
    System.out.println(rata[j]);
}
System.out.print("\nNilai Tertinggi\t");
for (int j = 0; j < banyakTes; j++) {
    System.out.print(ntt[j] + "\t");
}
System.out.println(rtt);

System.out.print("Nilai Terendah\t");
for (int j = 0; j < banyakTes; j++) {
    System.out.print(ntr[j] + "\t");
}
System.out.println(rtr);
}}

```

## TUGAS

### Tugas 1

-